Problem Sets 6

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To demonstrate the application of subset selection methods

Consider the "Hitters" data. For loading the data you need to install ISLR library. We wish to predict a baseball player's salary on the basis of various predictors. So first we need to choose an optimal model.

1. Clean the data by removing all rows that have missing value in any variable. You will be working with this clean data.

```
data(Hitters)
Hitters = na.omit(Hitters)
cat("Dimensions after cleaning:", dim(Hitters), "\n")

## Dimensions after cleaning: 263 20
```

2. Split the data randomly into train-test keeping 80% as training sample and remaining as test sample.

```
set.seed(123)
n = nrow(Hitters)
train_idx = sample(1:n, size = floor(0.8 * n))
train = Hitters[train_idx, ]
test = Hitters[-train_idx, ]
cat("Training size:", nrow(train), "| Test size:", nrow(test), "\n")

## Training size: 210 | Test size: 53
```

3. Perform the best subset selection on the training sample. While choosing the best model fit up to a 19 variable model. Choose the best model according to adjusted R squared, BIC, AIC and Cp values. In case the best model differs according to the selection criteria, report the best fitted model according to each selection criterion.

```
predict_regsubsets <- function(object, newdata, id) {
  form  = as.formula(object$call[[2]])
  mat   = model.matrix(form, newdata)
  coefi = coef(object, id = id)
  mat[, names(coefi)] %*% coefi
}

get_mse = function(object, traindata, testdata, nvars) {
  train_mat = model.matrix(Salary ~ ., traindata)
  test_mat  = model.matrix(Salary ~ ., testdata)

  sapply(nvars, function(i) {
```

```r
    coefi      <- coef(object, id = i)
    train_pred <- train_mat[, names(coefi)] %*% coefi
    test_pred  <- test_mat[, names(coefi)]  %*% coefi
    c(train_mse = mean((traindata$Salary - train_pred)^2),
      test_mse  = mean((testdata$Salary  - test_pred)^2))
  })
}

# Function to select best model per criterion and report MSEs
report_best <- function(reg_obj, train_df, test_df, method_name) {
  s <- summary(reg_obj)

  best_adjr2 <- which.max(s$adjr2)
  best_bic   <- which.min(s$bic)
  best_cp    <- which.min(s$cp)

  n_tr <- nrow(train_df)
  aic_vals <- n_tr * log(s$rss / n_tr) + 2 * (1:19 + 1)
  best_aic  <- which.min(aic_vals)

  criteria <- c(AdjR2 = best_adjr2, BIC = best_bic, AIC = best_aic, Cp =
best_cp)
  cat("\n===", method_name, "===\n")
  cat("Best model size by criterion:\n"); print(criteria)

  results <- data.frame(Criterion = names(criteria), Best_Size =
as.integer(criteria),
                        Train_MSE = NA, Test_MSE = NA)

  train_mat <- model.matrix(Salary ~ ., train_df)
  test_mat  <- model.matrix(Salary ~ ., test_df)

  for (i in seq_len(nrow(results))) {
    k     <- results$Best_Size[i]
    coefs <- coef(reg_obj, id = k)
    tr_p  <- train_mat[, names(coefs)] %*% coefs
    te_p  <- test_mat[, names(coefs)]  %*% coefs
    results$Train_MSE[i] <- mean((train_df$Salary - tr_p)^2)
    results$Test_MSE[i]  <- mean((test_df$Salary  - te_p)^2)
  }

  print(results)
  invisible(results)
}
```

3-4. Best Subset Selection

```
reg_best <- regsubsets(Salary ~ ., data = train, nvmax = 19, method =
"exhaustive")
res_best <- report_best(reg_best, train, test, "Best Subset Selection")
```

```
##
## === Best Subset Selection ===
## Best model size by criterion:
## AdjR2   BIC   AIC    Cp
##    12     5    12    11
##   Criterion Best_Size Train_MSE Test_MSE
## 1     AdjR2        12  88935.80 130066.9
## 2       BIC         5  97026.43 153090.5
## 3       AIC        12  88935.80 130066.9
## 4        Cp        11  89790.63 125652.3
```

5. Forward Stepwise Selection

```
reg_fwd <- regsubsets(Salary ~ ., data = train, nvmax = 19, method =
"forward")
res_fwd <- report_best(reg_fwd, train, test, "Forward Stepwise Selection")
```

```
##
## === Forward Stepwise Selection ===
## Best model size by criterion:
## AdjR2   BIC   AIC    Cp
##    14     7    14    11
##   Criterion Best_Size Train_MSE Test_MSE
## 1     AdjR2        14  88287.08 134001.7
## 2       BIC         7  95658.81 146540.9
## 3       AIC        14  88287.08 134001.7
## 4        Cp        11  90888.44 145923.9
```

6. Backward Stepwise Selection

```
reg_bwd <- regsubsets(Salary ~ ., data = train, nvmax = 19, method =
"backward")
res_bwd <- report_best(reg_bwd, train, test, "Backward Stepwise Selection")
```

```
##
## === Backward Stepwise Selection ===
## Best model size by criterion:
## AdjR2   BIC   AIC    Cp
##    13     5    13    13
##   Criterion Best_Size Train_MSE Test_MSE
## 1     AdjR2        13  88865.02 134736.7
## 2       BIC         5  97026.43 153090.5
## 3       AIC        13  88865.02 134736.7
## 4        Cp        13  88865.02 134736.7
```

7. Comparison of Three Methods

```r
cat("\n=== Summary Comparison ===\n")
```

```
##
## === Summary Comparison ===
```

```r
cat("\nBest Subset:\n");  print(res_best)
```

```
##
## Best Subset:
```

```
##   Criterion Best_Size Train_MSE Test_MSE
## 1     AdjR2        12  88935.80 130066.9
## 2       BIC         5  97026.43 153090.5
## 3       AIC        12  88935.80 130066.9
## 4        Cp        11  89790.63 125652.3
```

```r
cat("\nForward Stepwise:\n"); print(res_fwd)
```

```
##
## Forward Stepwise:
```

```
##   Criterion Best_Size Train_MSE Test_MSE
## 1     AdjR2        14  88287.08 134001.7
## 2       BIC         7  95658.81 146540.9
## 3       AIC        14  88287.08 134001.7
## 4        Cp        11  90888.44 145923.9
```

```r
cat("\nBackward Stepwise:\n"); print(res_bwd)
```

```
##
## Backward Stepwise:
```

```
##   Criterion Best_Size Train_MSE Test_MSE
## 1     AdjR2        13  88865.02 134736.7
## 2       BIC         5  97026.43 153090.5
## 3       AIC        13  88865.02 134736.7
## 4        Cp        13  88865.02 134736.7
```

Interpretation: All three methods tend to select similar models for most criteria (especially BIC and Cp which favor parsimony). Forward and Backward stepwise are computationally cheaper than Best Subset but may differ slightly in selected variables. Test MSE is the most reliable guide to generalization performance.